3.29 Fractional Ideals

Definition 3.29.1
Let A be an integral domain contained in its quotient field K. A non-zero A-submodule $I \subset K$ is called a ***fractional ideal*** *if there exists $0 \neq a \in A$ such that $aI \subset A$.*

We say a fractional ideal is ***principal*** *if it is generated by some $x \in K$ as an A-module.*

Proposition 3.29.2 (Fractional Ideals over a Noetherian Domain)
Let A be a Noetherian integral domain. Then every fractional ideal of A is a finitely generated A-module.

Proof. Let I be a fractional ideal and $0 \neq a \in A$ with $aI \subseteq A$. Then aI is an ideal of A, hence finitely generated, say by $x_1, ..., x_n$. Multiplication by a is an isomorphism $I \to aI$ of A-modules, so I is generated by $a^{-1}x_1, ..., a^{-1}x_n$. ∎

Proposition 3.29.3 (Operations on Fractional ideals)
Let A be an integral domain and $I, J \subset K$ fractional ideals. Then the following constructions are also fractional ideals

a) $I + J = \{i + j \mid i \in I j \in J\}$
b) $IJ = \left\{\sum_{k=1}^{n} i_k j_k \;\middle|\; i_k \in I j_k \in J\right\}$
c) $(I : J) = \{x \in K \mid xJ \subseteq I\}$

Proof. It is straightforward to prove that these constructions are again A-submodules of K.

Suppose that $aI \subseteq A$ and $bJ \subseteq A$, then $(ab)(I + J) \subseteq A$ and $(ab)IJ \subseteq A$. Similarly

$$(I : J)J \subseteq I \implies j(I : J) \subseteq I \implies abj(I : J) \subseteq A$$

for some $0 \neq j \in J$ ∎

Proposition 3.29.4
Let A be an integral domain. The collection of fractional ideals $\mathcal{I}_A$ forms a monoid under multiplication of ideals, with identity A.

Definition 3.29.5
Let A be an integral domain and I a fractional ideal. Then we say I is ***invertible*** *if there exists a fractional ideal J such that $IJ = A$.*

Proposition 3.29.6
Let A be an integral domain and I a fractional ideal. Then I is invertible if and only if $I(A : I) = A$, in which case $(A : I)$ is the unique inverse.

Proof. Uniqueness follows from general properties of monoids (...). If $I(A : I) = A$ then by definition I is invertible. Conversely suppose I is invertible, that is $IJ = A$. Then evidentally $J \subseteq (A : I)$ so $A = IJ \subseteq I(A : I) \subseteq A$. Therefore $(A : I)$ is an inverse for I as required. ∎

Corollary 3.29.7
Let A be an integral domain. Then the invertible fractional ideals form an abelian group under multiplication.

Definition 3.29.8 (Localisation of Fractional Ideals)
Let A be an integral domain contained in its quotient field K. Let I be a fractional ideal and S a multiplicatively closed subset of A. Then define

$$S^{-1}I \coloneqq \left\{\frac{i}{s} \;\middle|\; i \in I s \in S\right\}$$

If $\mathfrak{a}$ is an ideal then $S^{-1}\mathfrak{a}$ corresponds to the usual notion (3.5.17) and is equal to the extension of $\mathfrak{a}$ under the inclusion homomorphism $A \to S^{-1}A$.

If $\mathfrak{p}$ is a prime ideal of A then as usual define

$$I_{\mathfrak{p}} \coloneqq (A \setminus \mathfrak{p})^{-1} I$$

and we may write

$$\mathfrak{a}_{\mathfrak{p}} = \mathfrak{a} A_{\mathfrak{p}}$$

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Proposition 3.29.9 (Localisation of Fractional Ideals)
Let A be an integral domain contained in its quotient field K. Let S be a multiplicatively closed subset of A and I a fractional ideal. Then regarding as subrings of K then $S^{-1}I$ is a fractional ideal of $S^{-1}A$. Furthermore

a) $S^{-1}(I+J) = S^{-1}I + S^{-1}J$
b) $S^{-1}(IJ) = (S^{-1}I)(S^{-1}J)$
c) $S^{-1}(I:J) = (S^{-1}I : S^{-1}J)$ *if J is finitely generated (in general only $\subseteq$ holds)*

Proof. Evidently $S^{-1}I$ is again a fractional ideal of $S^{-1}A$.

a) Clear because $\frac{i+j}{s} = \frac{i}{s} + \frac{j}{s}$.
b) We have $\frac{ij}{s} = \left(\frac{i}{s}\right)\left(\frac{j}{1}\right)$ whence $S^{-1}(IJ) \subseteq (S^{-1}I)(S^{-1}J)$. Conversely we have $\left(\frac{i}{s}\right)\left(\frac{j}{t}\right) = \frac{ij}{st}$ so we ahve the reverse inclusion.
c) Suppose that $\frac{x}{s} \in S^{-1}(I:J)$ and $\frac{j}{t} \in S^{-1}J$. Then $\left(\frac{x}{s}\right)\left(\frac{j}{t}\right) = \frac{xj}{st} \in S^{-1}I$ because $x \in (I:J)$. Therefore $S^{-1}(I:J) \subseteq (S^{-1}I : S^{-1}J)$. Conversely suppose $J = (j_1, ..., j_n)$ and let $x \in (S^{-1}I : S^{-1}J)$. Then for each k we have $x\left(\frac{j_k}{1}\right) = \frac{i_k}{s_k}$ with $i_k \in I$ and $s_k \in S$. Put $s := s_1 ... s_n \in S$, so $sxj_k \in I$ for every k, whence $sxJ \subseteq I$ as the j_k generate J, that is $sx \in (I:J)$. Therefore $x = \frac{sx}{s} \in S^{-1}(I:J)$ as required. (Finite generation is what lets a single s serve all of J.)

■

Lemma 3.29.10
Let A be an integral domain contained in its quotient field K. Suppose I is a fractional ideal then

$$I = \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$$

In particular we have

$$A = \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$$

Proof. Evidently $I \subseteq \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Suppose that $x \in \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Consider the ideal $\mathfrak{a} := \{a \in A \mid ax \in I\}$. Suppose $\mathfrak{a}$ is proper then by (3.3.15) it is contained in a maximal ideal $\mathfrak{m}$. As $x \in I_{\mathfrak{m}}$ we have $(A \setminus \mathfrak{m}) \cap \mathfrak{a} \neq \emptyset$ and $\mathfrak{a} \subseteq \mathfrak{m}$ a contradiction. Therefore $\mathfrak{a} = A$ and in particular $x = 1.x \in I$ as required. ■

Proposition 3.29.11 (Invertibility is a local property)
Let A be an integral domain with quotient field K, and I a fractional ideal of A. Consider the conditions

a) *I is invertible*
b) *$I_{\mathfrak{p}}$ is invertible for all prime ideals $\mathfrak{p}$*
c) *$I_{\mathfrak{m}}$ is invertible for all maximal ideals $\mathfrak{m}$*

Then a) $\implies$ b) $\implies$ c), and if I is finitely generated then c) $\implies$ a). In particular all three are equivalent when A is Noetherian, as then every fractional ideal is finitely generated (3.29.2).

Proof. a) $\implies$ b) Let $IJ = A$ for a fractional ideal J. By (3.29.9) $I_{\mathfrak{p}}J_{\mathfrak{p}} = (IJ)_{\mathfrak{p}} = A_{\mathfrak{p}}$, so $I_{\mathfrak{p}}$ is invertible with inverse $J_{\mathfrak{p}}$.

b) $\implies$ c) Every maximal ideal is prime (3.3.60) so we are done.

c) $\implies$ a) Suppose that I is finitely generated. Then

$$\begin{aligned}
I(A:I) &= \bigcap_{\mathfrak{m}} (I(A:I))_{\mathfrak{m}} && (3.29.10)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A:I)_{\mathfrak{m}} && (3.29.9)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A_{\mathfrak{m}} : I_{\mathfrak{m}}) && (3.29.9)\\
&= \bigcap_{\mathfrak{m}} A_{\mathfrak{m}} && (3.29.6)\\
&= A && (3.29.10)
\end{aligned}$$

where the third equality uses that I is finitely generated, and the second last that $I_{\mathfrak{m}}$ is invertible by assumption. ■

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